

Ex. Nos. 5	APPLYING FILTERING AND PROJECTIONS
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Aim:

To perform an filtering and projection operation on Vehicles collection.

PROCEDURE:

A. Insert Data

```
db.students.insertMany([
  { name: "Mukesh", age: 19, dept: "IT", marks: 85 },
  { name: "Arun", age: 20, dept: "CSE", marks: 75 },
  { name: "Priya", age: 19, dept: "IT", marks: 92 },
  { name: "Kiran", age: 21, dept: "ECE", marks: 65 }
])
```

♦ **Filtering**

```
db.students.find({ dept: "IT" })
```

♦ **Output**

```
[
  { "name": "Mukesh", "dept": "IT", "marks": 85 },
  { "name": "Priya", "dept": "IT", "marks": 92 }
]
```

♦ **Projection**

```
db.students.find(
  { marks: { $gt: 80 } },
  { name: 1, marks: 1, _id: 0 }
)
```

♦ **Output**

```
[
  { "name": "Mukesh", "marks": 85 },
  { "name": "Priya", "marks": 92 }
]
```

B Insert Data

```
db.products.insertMany([
  { product: "Laptop", price: 60000, stock: 10, category: "Electronics" },
  { product: "Mouse", price: 500, stock: 50, category: "Electronics" },
  { product: "Chair", price: 2000, stock: 15, category: "Furniture" },
  { product: "Table", price: 5000, stock: 5, category: "Furniture" }
])
```

♦ Filtering

```
db.products.find({ category: "Electronics" })
```

♦ Output

```
[
  { "product": "Laptop", "category": "Electronics", "price": 60000 },
  { "product": "Mouse", "category": "Electronics", "price": 500 }
]
```

♦ Projection

```
db.products.find(
  { stock: { $lt: 20 } },
  { product: 1, stock: 1, _id: 0 }
)
```

♦ Output

```
[
  { "product": "Laptop", "stock": 10 },
  { "product": "Chair", "stock": 15 },
  { "product": "Table", "stock": 5 }
]
```

Result:

Thus, the queries to perform various filtering and Projection operations on students and .products collection is executed successfully